s6_neutral_levered

equity strategy | family: equity-neutralized | status: validated | derived from `s5_sector_neutral`

What this strategy does

Strategy 5 weights, 2.5x gross leverage cap, base unit 0.003

Why it should work. Market-neutral earnings drift with enough gross exposure to matter; the neutrality is the hedge.

Design notes (from the strategy's source file).

Strategy 6 -- Sector/size-neutral book with the leverage cap raised to 2.5x.

Strategy 5's weights, but with the 1.5x cap lifted to 2.5x (the point where it stops binding) and sizing retuned. Because sector+size neutralization already prevents concentration, the cap was protecting nothing: return rises 4.1% -> 6.5% with the Sharpe holding at ~1.3. The pure-alpha reference strategy for the repo.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

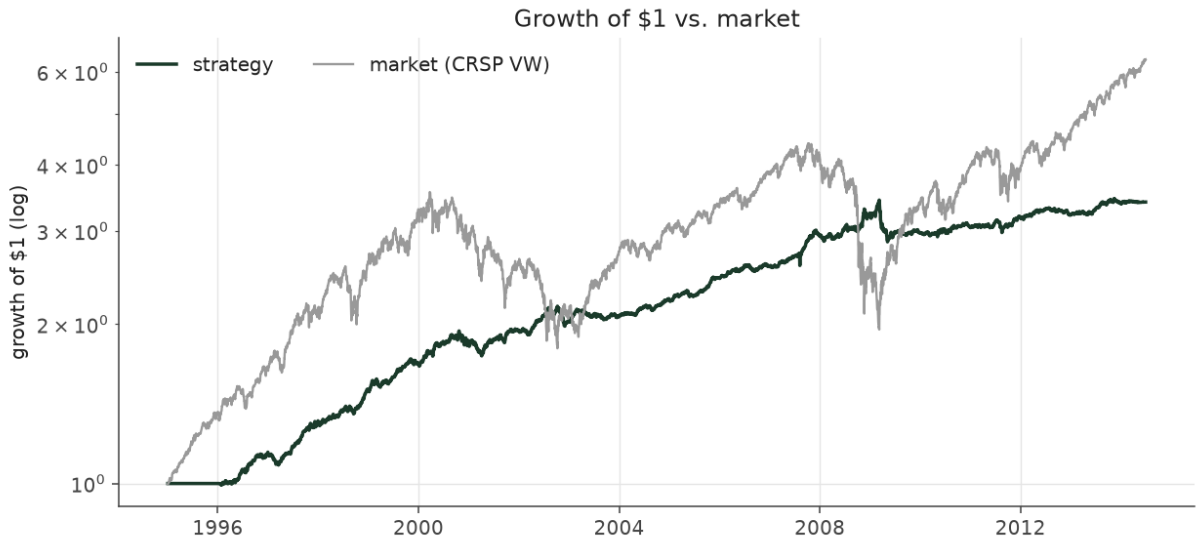
metric	s6 neutral levered	market (CRSP VW)
Annualized return	6.49%	9.94%
Annualized volatility	4.94%	19.35%
Sharpe (NAV-pct)	1.30	0.59
Sortino	1.90	
Max drawdown	-16.6%	-55.5%
Calmar	0.39	
Final NAV ($10M start)	$34,048,037	
Worst day	-3.57%	
% positive days	52.0%	
Sharpe 1996-2001	2.07	
Sharpe 2002-2007	1.52	
Sharpe 2008-2013	0.47	

Correlation of daily returns with the market: **0.02**.

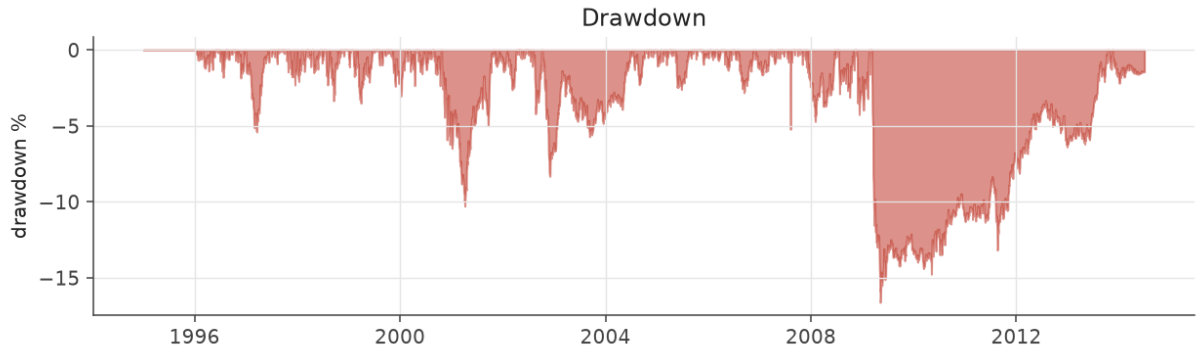
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 1.3

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0649

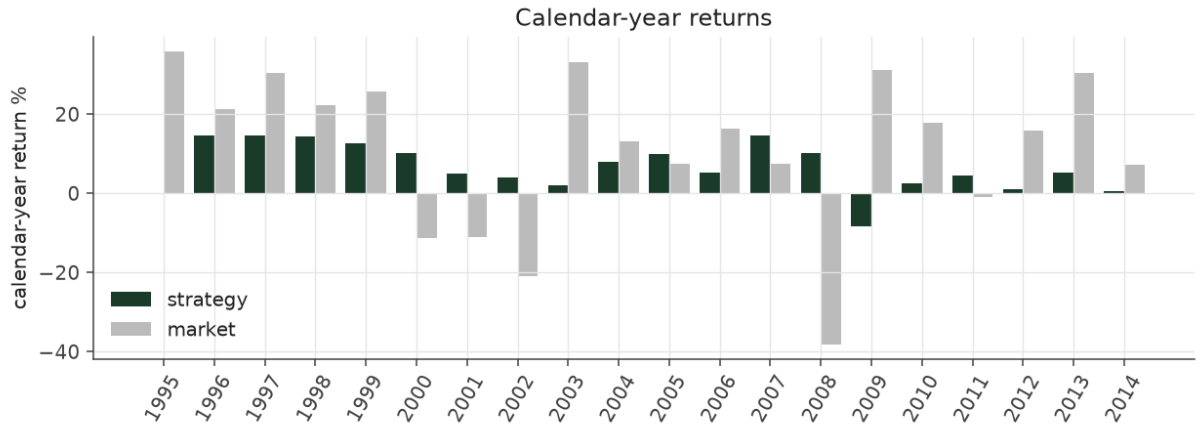
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.391



NAV (log scale)



Drawdown from running peak

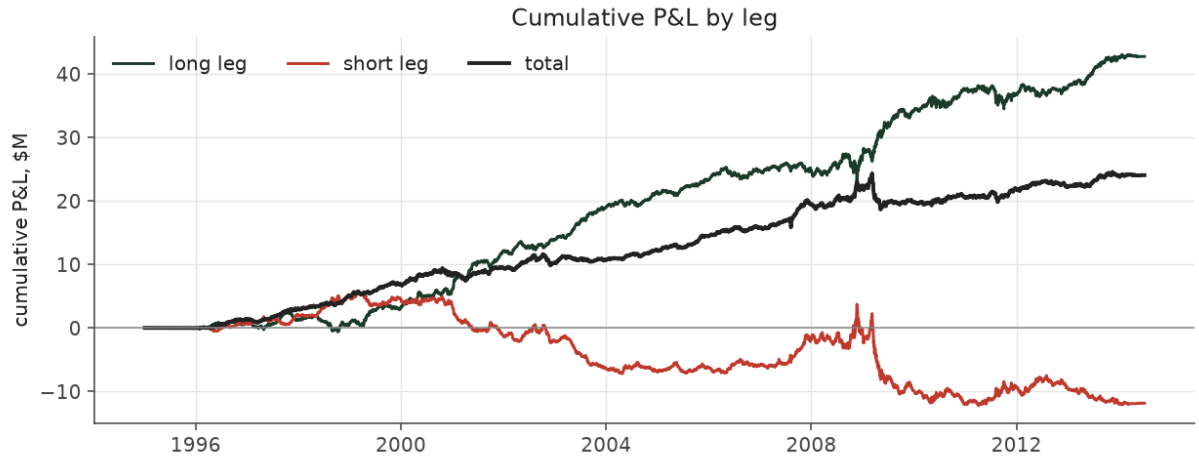


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	42.74	-11.85
P&L after costs ($M)	39.45	-15.40
of which common (average-event) component ($M)	26.88	-27.72
of which excess / PEAD component ($M)	15.86	15.87
excess minus costs ($M)	12.57	12.32
gross return per $ traded (bp)	253.36	-68.44
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.22	-0.24

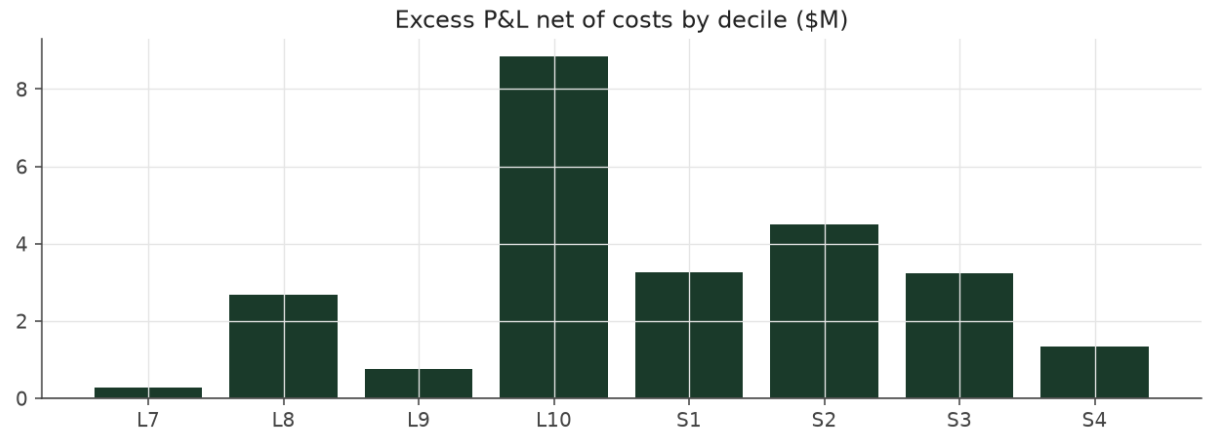
Daily P&L correlation between the two legs: **-0.81**.

• **The short leg loses money in absolute terms** (-11.8M before costs) while the long leg earns +42.7M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 26.9M / 27.7M. Stripping that common component, the surprise signal itself earned +15.9M on the long side and +15.9M on the short side.

• The two legs' daily P&L are -0.81 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.22 long / -0.24 short vs 1.30 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	148.61	3.03	0.56	0.29	0.27	1.1%	18.00
long	8	16,588	394.76	9.19	3.47	0.78	2.69	10.8%	68.11
long	9	17,747	522.21	8.37	1.77	1.01	0.76	3.1%	14.60
long	10	17,174	621.45	18.85	10.07	1.22	8.85	35.6%	142.39
short	1	17,169	624.20	-6.03	4.58	1.33	3.25	13.1%	52.06
short	2	17,281	539.87	-4.65	5.58	1.09	4.49	18.0%	83.16
short	3	15,970	404.59	-3.39	4.05	0.81	3.24	13.0%	79.97
short	4	9,171	162.59	-1.33	1.66	0.31	1.35	5.4%	82.78



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	306.96	5.45	7.41	1.18	6.24	25.1%	203.15
2	22,885	641.27	6.15	8.35	1.80	6.55	26.3%	102.10
3	25,501	785.23	6.22	7.75	1.64	6.11	24.6%	77.86
4	27,583	839.10	3.71	4.93	1.31	3.62	14.5%	43.15
5	28,947	845.72	2.51	3.30	0.93	2.37	9.5%	28.03

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	225.40	5.20	6.74	1.13	5.61	22.6%	249.07
2	23,084	676.60	5.79	8.72	2.03	6.69	26.9%	98.85
3	25,851	834.81	6.30	8.50	1.67	6.83	27.5%	81.85
4	27,665	843.84	4.54	5.11	1.18	3.93	15.8%	46.53

5	28,874	837.44	2.21	2.66	0.84	1.82	7.3%	21.71

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	1,296.80	-0.36	7.02	2.28	4.74	19.1%	36.57
41-60d	25,652	773.42	4.67	4.86	1.58	3.28	13.2%	42.46
61-90d	35,324	1,006.66	7.13	8.63	2.12	6.51	26.2%	64.70
>90d	12,298	341.40	12.61	11.22	0.87	10.35	41.6%	303.13

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	702.89	9.20	10.91	1.38	9.54	38.3%	135.71
2002-07	39,956	1,131.98	10.13	12.35	2.22	10.13	40.7%	89.47
2008-13	45,278	1,583.40	4.72	8.47	3.25	5.22	21.0%	32.97

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	335.42	1.77	2.48	0.71	1.77	7.1%	52.89
Chems	2,590	227.57	1.44	1.85	0.40	1.45	5.8%	63.75
Durbl	2,843	245.81	2.39	2.91	0.45	2.46	9.9%	100.13
Enrgy	6,194	278.68	3.95	4.57	0.52	4.05	16.3%	145.38
Hlth	11,696	325.36	2.67	3.38	0.68	2.69	10.8%	82.79
Manuf	12,034	304.84	1.76	2.55	0.63	1.92	7.7%	62.91
Money	17,583	299.07	2.59	3.36	0.67	2.69	10.8%	89.95
NoDur	5,081	274.80	1.38	1.87	0.52	1.35	5.4%	49.05
Other	18,377	325.08	2.31	3.13	0.72	2.41	9.7%	74.15
Shops	11,984	317.62	3.13	3.82	0.65	3.17	12.7%	99.77
Telcm	3,981	270.57	0.18	0.75	0.49	0.26	1.0%	9.46
Utils	4,672	213.45	0.48	1.06	0.39	0.67	2.7%	31.19

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	1,094.31	4.11	7.37	1.98	5.39	21.7%	49.25
low vol	40,091	1,165.68	10.19	12.82	2.56	10.26	41.2%	87.97
mid vol	37,903	1,105.61	9.59	11.28	2.19	9.09	36.5%	82.20

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	6.49%	0.00%
0.5%	1.8	6.20%	0.29%
1.0%	3.5	5.90%	0.59%
2.0%	7.0	5.24%	1.25%
5.0%	17.5	2.61%	3.88%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.